

MATTHEW CHOO

Toronto, Ontario

☎ 416-838-1801 ✉ mmakchoo@gmail.com [linkedin.com/in/matthew-choo](https://www.linkedin.com/in/matthew-choo)

Education

University of Western Ontario

Bachelor of Science, Mechanical Engineering

Sep. 2024 – May 2028

London, ON

Experience

BFC Technologies Inc.

Process/Mechanical Engineering Intern

Jun 2025 – Aug 2025

Woodbridge, ON

- Supported mechanical/process design for pharmaceutical process super skids; blended QA/QC, SolidWorks, and GMP documentation to deliver turn-key systems.
- Collaborated with pharmaceutical clients (Eli Lilly, Fluor, United Therapeutics, Amgen, CRB, Jacobs) and vendors (Inox, CFF, Pinnacle) on process skid design, QA/QC, and FAT/SAT execution.
- Verified spool drawings, weld logs, and tags; led skid walkdowns; executed FAT/SATs and compiled turnover packages (P&IDs, weld logs, BOMs, MTRs, calibration certs).
- Designed assemblies and pipe supports in SolidWorks; specified bolting for gasketed joints per ASME PCC-1 and B16.5.
- Performed calculations for blowdown systems, valve actuation, and compressed-air headers, applying ASME BPE hygienic design.
- Managed POs, vendor coordination, and tracking for large-scale flanged and tri-clamp assemblies.
- Estimated skid costs from P&IDs and hardware take-offs; supported bids with pricing and scope breakdowns.
- Applied standards: ASME PCC-1, ASME B16.5, ASME BPVC VIII, ASME BPE, ASME B31.3, ISO/IEC 17025, FDA cGMP, Health Canada GMP.

Various Companies

Swim/Skiing Instructor (Contract Part-Time)

Dec 2021 – Sept 2024

Richmond Hill, ON

- Taught swimming and skiing combining safety-first instruction, clear communication, and adaptive coaching.

Technical Skills

Engineering/Standards: FEA, CFD, P&ID's, DFMA, GMP, ASME, ISO/IEC 17025, SAE J2966

Design: SolidWorks, Inventor, Autocad, Navisworks, Bluebeam, OnShape, Fusion 360, Arduino IDE

Software: SolidWorks, Inventor, Autocad, Navisworks, Bluebeam, OnShape, Fusion 360, Excel, Word, PowerPoint

Engineering/Standards:

Programming: Java, Matlab, Arduino IDE.

Projects

Baja SAE Aerodynamics Project | *SolidWorks, Inventor, Ansys, FEA, CFD, Prototyping, SAE J2966*

Sept 2025–present

- Redesigned vehicle panels to reduce coefficient of drag and achieve 1–2.5 kph top speed increase
- Developed parametric CAD for CFD analysis under multiple yaw angles; validated design for ground effect and wake interactions.
- Managed primary risks: performance gain uncertainty, serviceability, rule compliance, manufacturing complexity.
- Ensured upper frame wrapping, E-stop access, number plate specs, CVT cooling line routing, refueling and electrical serviceability per SAE specifications.
- Researched mounting systems, panel configurations, integration strategies, and SAE rules; coordinated with cross-functional teams and led testing planning.

Throttle System – Baja SAE | *SolidWorks, FEA, Prototyping, User Design*

Sept 2024–April 2025

- Refit throttle system for new vehicle frame using solid modeling and FEA to optimize strength and reduce weight, ensuring design met Baja SAE rules.
- Fabricated 3D printed prototypes, coordinated with Brake Capstone team to ensure compatibility and performance.

Extracurricular

Baja SAE

Business Lead

Jun 2025 – Current

University of Western Ontario

- Led the business and operations division for the Baja SAE team after transitioning from technical Drivetrain role.
- Organized sponsor relations, fundraising initiatives, and budget management to secure resources for competition.
- Coordinated team logistics, event planning, and project presentations to judges and stakeholders.
- Guided business case development and refined strategic messaging, improving team visibility within engineering faculty.

Baja SAE

Drivetrain Member

Sept 2024 – May 2025

University of Western Ontario

- Applied engineering principles in off-road vehicle design and build for competition; utilized CAD and FEA for component design.
- Observed project planning, scheduling, documentation during team sessions.

Western Engineering

Competitor

Oct 2024 – Dec 2025

University of Western Ontario

- Western Engineering Open House, Western Engineering 3rd place Senior Design